Six-Month Research Progress Report

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1. Research Overview

My doctoral research focuses on ordinal-pattern (Bandt–Pompe) information-theoretic features, including permutation, Rényi and Tsallis entropies, Fisher information, and statistical complexity, for analyzing and clustering geophysical time series. Over the past six months, I have applied these features to two real-world problems, synthetic aperture radar (SAR) change detection and volcanic seismic monitoring at Whakaari/White Island, while advancing the simulation and writing components of my thesis.

2. Contribution to the IEEE MIGARSS-InGARSS 2026 Conference Paper (SAR)

We submitted a conference paper on ordinal patterns for SAR change detection for the 2026 IEEE MIGARSS-InGARSS conference, in collaboration with researchers at IIT Indore, and my secondary supervisor is from Universidad Nacional de Hurlingham. Paper status: **Accepted**. My contributions were:

- Building the R pipeline that encodes Sentinel-1 SAR pixel time series and computes ordinal-pattern features (Shannon entropy, statistical complexity, and Fisher information).
- Producing the entropy–complexity ($H \times C$) plane analysis of change and no-change pixels at embedding dimensions $D = 3$, 4, and 5, which showed that $D = 3$ leads to entropy saturation and supported the recommendation of $D = 4$ or 5 for series of length 740.
- Preparing the figures and typesetting the manuscript in the IEEE template (Quarto/LaTeX, Ti*k*Z).

3. Seismic Amplitude Research at Whakaari/White Island

Using seismic amplitude data from the WIZ and WSRZ stations, I built a reproducible R workflow covering data retrieval from the university server, preprocessing, feature computation and visualisation.

- Computed weekly ordinal-pattern features (7-day windows with 1-day overlap and 6-day overlap) from 10-minute RSAM data for July 2014 – December 2019, for the entropy and complexity measures.
- Applied k-means and DBSCAN clustering and a class-weighted RBF support vector machine, and validated the results against the GeoNet Volcanic Alert Level (VAL) record using PCA and Sankey visualizations.
- Found that clustering alone did not reliably isolate eruptive periods, such as the 9 December 2019 eruption, even though these periods are clearly visible in the $H \times C$ plane and feature time series. This led me to develop a threshold-based anomaly detection approach and an event coincidence analysis (ECA) that tests whether feature-anomaly weeks coincide with eruption weeks, a novel application of ordinal-pattern features.
- Drafted a report, "Ordinal-Pattern Cluster Analysis of RSAM Time Series for Detecting Volcanic Seismic Anomalies at Whakaari, 2014–2019", and shared the results with a volcanologist collaborator for expert review of the ground-truth labels.

4. Thesis Development

I have largely completed the literature review, which surveys how ordinal-pattern features are used across scientific domains and identifies a clear gap: these features have rarely been used for time series clustering or event coincidence analysis. I have also produced simulation results in which 13 ordinal-pattern features computed from convex-combination time series are clustered using the $H \times C$ plane, k-means and DBSCAN. I am currently writing the methodology chapter for completed work.

5. Plans for the Next Six Months

I plan to finalize the ECA and threshold-based anomaly detection for Whakaari and prepare the results as a journal manuscript; complete the Wayanad landslide study; integrate the simulation results into the methodology chapter; and complete a full draft of all thesis chapters by April 2027. Continued scholarship support would allow me to complete these outputs and submit my thesis on schedule.